

Exercise 18.5

1. Find the second order derivative of the following.

a) x^{20}

let $y = x^{20}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d x^{20}}{dx}$$

$$= 20x^{19}$$

differentiating again with respect to x .

$$\frac{d^2 y}{dx^2} = \frac{d 20x^{19}}{dx}$$

$$= 20 \times 19 x^{18}$$

$$= 380 x^{18} \quad \checkmark$$

b) $x^3 - 3x^2 + 2x - 1$

let $y = x^3 - 3x^2 + 2x - 1$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d(x^3 - 3x^2 + 2x - 1)}{dx}$$

$$\frac{dy}{dx} = 3x^2 - 6x + 2$$

differentiating again with respect to x .

$$\frac{d^2 y}{dx^2} = \frac{d(3x^2 - 6x + 2)}{dx}$$

$$= 6x - 6 \quad \checkmark$$

c) $\frac{3}{x^2}$

→ let $y = \frac{3}{x^2}$

differentiating with respect to x .

$$\frac{dy}{dx} = \frac{d\left(\frac{3}{x^2}\right)}{dx}$$

$$= 3 \frac{d x^{-2}}{dx}$$

$$= 3 \times -2 x^{-3}$$

$$= -6 x^{-3}$$

$$= \underline{\underline{\frac{-6}{x^3}}}$$

differentiating again with respect to x .

$$\frac{d^2 y}{dx^2} = \frac{d(-6 x^{-3})}{dx}$$

$$= -6 \frac{d x^{-3}}{dx}$$

$$= -6 \times -3 x^{-4}$$

$$= \underline{\underline{\frac{18}{x^4}}}$$

d) $\frac{1}{\sqrt{3x}}$

→ let $y = \frac{1}{\sqrt{3x}}$

$$= \frac{1}{\sqrt{3} x^{1/2}}$$

$$= \frac{1}{\sqrt{3}} \times x^{-1/2}$$

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differentiating with respect to x

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\sqrt{3}} \frac{d x^{-1/2}}{dx} \\&= \frac{1}{\sqrt{3}} \times -\frac{1}{2} x^{-1/2-1} \\&= \frac{-1}{2\sqrt{3}} x^{-3/2}\end{aligned}$$

differentiating again with respect to x .

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{-1}{2\sqrt{3}} \frac{d x^{-3/2}}{dx} \\&= \frac{-1}{2\sqrt{3}} \times -\frac{3}{2} x^{-3/2-1} \\&= \frac{\sqrt{3}}{4} x^{-5/2}\end{aligned}$$

e) $\sqrt{x^2+2}$

→ let $y = \sqrt{x^2+2}$
 $y = (x^2+2)^{1/2}$

differentiating with respect to x .

$$\begin{aligned}\frac{dy}{dx} &= \frac{d(x^2+2)^{1/2}}{dx} \\&= \frac{d(x^2+2)^{1/2}}{d(x^2+2)} \times \frac{d(x^2+2)}{dx} \\&= \frac{1}{2} (x^2+2)^{-1/2} \times \left(\frac{dx^2}{dx} + \frac{d2}{dx} \right) \\&= \frac{1}{2} (x^2+2)^{-1/2} \times 2x \\&= x (x^2+2)^{-1/2}\end{aligned}$$

differentiate again with respect to x .

$$\begin{aligned}
 \frac{d^2 y}{dx^2} &= \frac{d}{dx} [x(x^2+2)^{-1/2}] \\
 &= x \frac{d(x^2+2)^{-1/2}}{d(x^2+2)} \times \frac{d(x^2+2)}{dx} + (x^2+2)^{-1/2} \times \frac{dx}{dx} \\
 &= x \left[\frac{-1}{2} (x^2+2)^{-3/2} \times 2x \right] + (x^2+2)^{-1/2} \\
 &= -x^2 (x^2+2)^{-3/2} + (x^2+2)^{-1/2} \\
 &= \frac{1}{(x^2+2)^{1/2}} - \frac{x^2}{(x^2+2)^{3/2}} \\
 &= \frac{x^2+2-x^2}{(x^2+2)^{3/2}} \\
 &= \frac{2}{(x^2+2)^{3/2}}
 \end{aligned}$$

2. If $y = 5 \cos x - 3 \sin x$, prove that $\frac{d^2 y}{dx^2} + y = 0$

$$y = 5 \cos x - 3 \sin x$$

differentiating with respect to x .

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d(5 \cos x - 3 \sin x)}{dx} \\
 &= \frac{d 5 \cos x}{dx} - \frac{d 3 \sin x}{dx} \\
 &= 5(-\sin x) - 3 \cos x \\
 &= -5 \sin x - 3 \cos x
 \end{aligned}$$

differentiating again with respect to x .

$$\begin{aligned}
 \frac{d^2 y}{dx^2} &= \frac{d(-5 \sin x - 3 \cos x)}{dx} \\
 &= -5 \cos x - 3(-\sin x) \\
 &= -5 \cos x + 3 \sin x
 \end{aligned}$$

Now,

$$\frac{d^2y}{dx^2} + y = -5\cos x + 3\sin x + 5\cos x - 3\sin x$$

$$= 0$$

$$\therefore \frac{d^2y}{dx^2} + y = 0$$

proved

3. Find the value of $f''(x)$ of the function $f(x) = x^3 + x^2 - 3x$ at the point $x=2$

$$\rightarrow f(x) = x^3 + x^2 - 3x$$

$$f'(x) = 3x^2 + 2x - 3$$

$$f''(x) = 6x + 2$$

Now,

$$f''(2) = 6 \times 2 + 2$$

$$= 12 + 2$$

$$= 14$$

$$\therefore f''(2) = 14$$